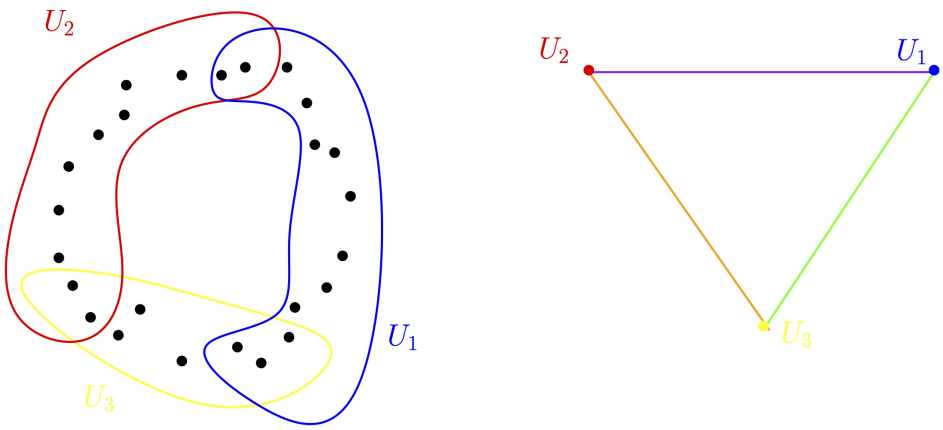

Figure 3: The nerve of a cover of a set of sampled points in the plane.

of continuous spaces into abstract combinatorial structures that are well-suited for the design of effective data structures and algorithms.

3 Using covers and nerves for exploratory data analysis and visualization: the Mapper algorithm

Using the nerve of covers as a way to summarize, visualize and explore data is a natural idea that was first proposed for TDA in Singh et al. (2007), giving rise to the so-called Mapper algorithm.

Definition 2. *Let $f : X \to \mathbb{R}^d$, $d \geq 1$, be a continuous real valued function and let $\mathcal{U} = (U_i)_{i \in I}$ be a cover of $\mathbb{R}^d$. The pull back cover of X induced by $(f, \mathcal{U})$ is the collection of open sets $(f^{-1}(U_i))_{i \in I}$. The refined pull back is the collection of connected components of the open sets $f^{-1}(U_i)$, $i \in I$.*

The idea of the Mapper algorithm is, given a data set $\mathbb{X}$ and well-chosen real valued function $f : \mathbb{X} \to \mathbb{R}^d$, to summarize $\mathbb{X}$ through the nerve of the refined pull back of a cover $\mathcal{U}$ of $f(\mathbb{X})$, see Figure 4(a) . For well-chosen covers $\mathcal{U}$ (see below), this nerve is a graph providing an easy and convenient way to visualize the summary of the data. It is described in Algorithm 1 and illustrated on a simple example in Figure 4(b).

The Mapper algorithm is very simple but it raises several questions about the various choices that are left to the user and that we briefly discuss in the following.

Algorithm 1 The Mapper algorithm

Input: A data set $\mathbb{X}$ with a metric or a dissimilarity measure between data points, a function $f : \mathbb{X} \to \mathbb{R}$ (or $\mathbb{R}^d$), and a cover $\mathcal{U}$ of $f(\mathbb{X})$.
for each $U \in \mathcal{U}$, decompose $f^{-1}(U)$ into clusters $C_{U,1}, \cdots, C_{U,k_U}$.
Compute the nerve of the cover of X defined by the $C_{U,1}, \cdots, C_{U,k_U}$, $U \in \mathcal{U}$
Output: a simplicial complex, the nerve (often a graph for well-chosen covers $\to$ easy to visualize):
- a vertex $v_{U,i}$ for each cluster $C_{U,i}$,
- an edge between $v_{U,i}$ and $v_{U',j}$ iff $C_{U,i} \cap C_{U',j} \neq \emptyset$

The choice of f. The choice of the function f, sometimes called the filter or lens function, strongly depends on the features of the data that one expect to highlight. The following ones are among the ones more or less classically encountered in the literature:

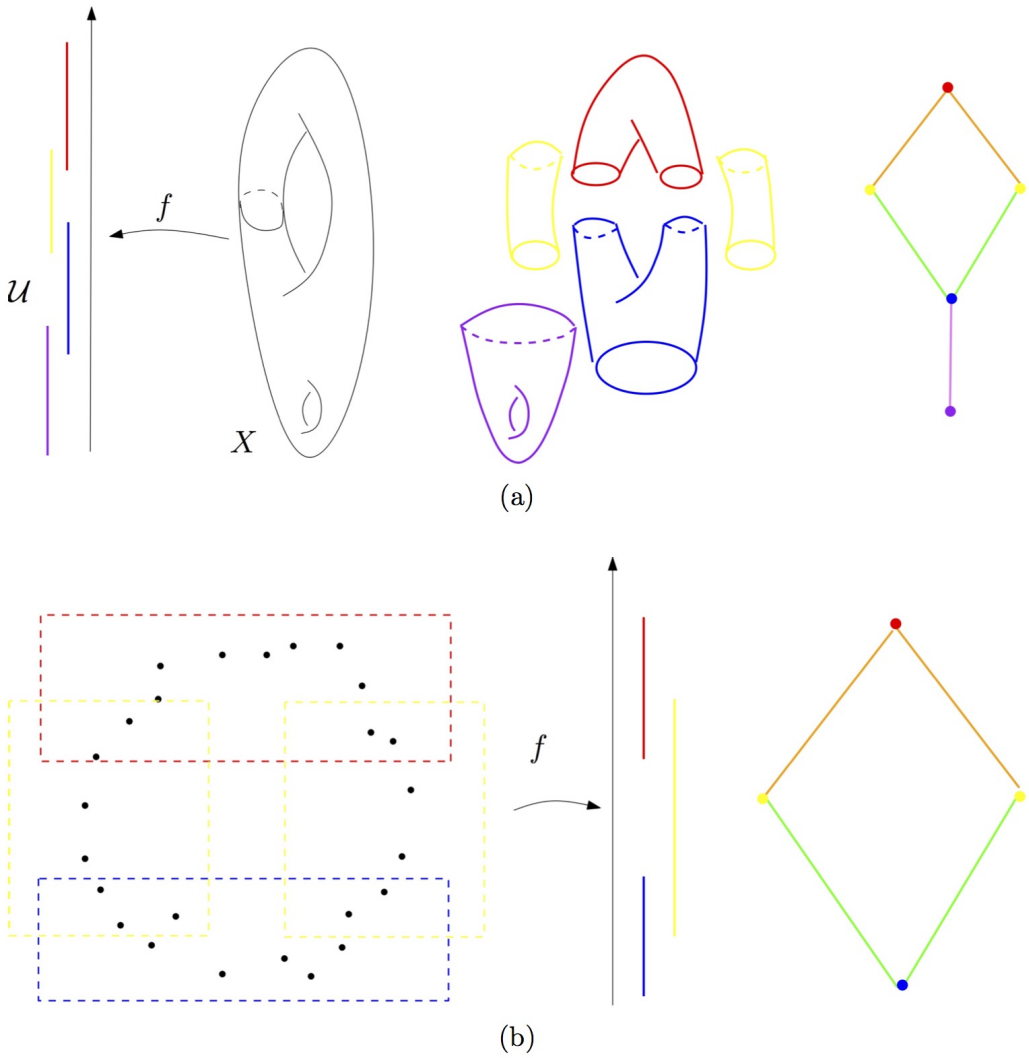

Figure 4: (a) The refined pull back cover of the height function on a surface in $\mathbb{R}^3$ and its nerve. (b) The mapper algorithm on a point cloud sampled around a circle.

- Density estimates: the mapper complex may help to understand the structure and connectivity of high density areas (clusters).
- PCA coordinates or coordinates functions obtained from a non linear dimensionality reduction (NLDR) technique, eigenfunctions of graph laplacians,... may help to reveal and understand some ambiguity in the use of non linear dimensionality reductions.
- The centrality function $f(x) = \sum_{y \in \mathbb{X}} d(x, y)$ and the eccentricity function $f(x) = \max_{y \in \mathbb{X}} d(x, y)$, appears sometimes to be good choices that do not require any specific knowledge about the data.
- For data that are sampled around 1-dimensional filamentary structures, the distance function to a given point allows to recover the underlying topology of the filamentary structures Chazal et al. (2015c).

The choice of the cover $\mathcal{U}$. When f is a real valued function, a standard choice is to take $\mathcal{U}$ to be a set of regularly spaced intervals of equal length $r > 0$ covering the set $f(\mathbb{X})$. The real r is sometimes called the *resolution* of the cover and the percentage g of overlap between two consecutive intervals is called the the *gain* of the cover. Note that if the gain g is chosen below 50%, then every point of the real line is covered by at most 2 open sets of $\mathcal{U}$ and the output nerve is a graph. It is important to notice that the output of the Mapper is very sensitive to the choice of $\mathcal{U}$ and small changes in the resolution and gain parameters may results in very large changes in the output, making the method very unstable. A classical strategy consists in exploring some range of parameters and select the ones that turn out to provide the most informative output from the user perspective.

The choice of the clusters. The Mapper algorithm requires to cluster the preimage of the open sets $U \in \mathcal{U}$. There are two strategies to compute the clusters. A first strategy consists in applying, for each $U \in \mathcal{U}$, a cluster algorithm, chosen by the user, to the premimage $f^{-1}(U)$. A second, more global, strategy consists in building a neighboring graph on top of the data set $\mathbb{X}$, e.g. k-NN graph or ε-graph, and, for each $U \in \mathcal{U}$, taking the connected components of the subgraph with vertex set $f^{-1}(U)$.

Theoretical and statistical aspects of Mapper. Based on the results on stability and the structure of Mapper proposed in Carrière and Oudot (2015), advances towards a statistically well-founded version of Mapper have been obtained recently in Carriere et al. (2018). Unsurprisingly, the convergence of Mapper depends on both the sampling of the data and the regularity of the filter function. Moreover, subsampling strategies can be proposed to select a complex in a Rips filtration at a convenient scale, as well as the resolution and the gain for defining the Mapper graph. The case of stochastic and multivariate filters has also been studied in Carrière and Michel (2019). An alternative description of the probabilistic convergence of Mapper, in term of categorification, has also been proposed in Brown et al. (2020). Other approaches have been proposed to study and deal with the instabilities of the Mapper algorithm in Dey et al. (2016, 2017).

Data Analysis with Mapper. As an exploratory data analysis tool, Mapper has been successfully used for clustering and feature selection. The idea is to identify specific structures in the Mapper graph (or complex), in particular loops and flares. These structures are then used to identify interesting clusters or to select features or variable that best discriminate the data in these structures. Applications on real data, illustrating these techniques, may be found, for example, in Carrière and Rabadán (2020); Lum et al. (2013); Yao et al. (2009).

4 Geometric reconstruction and homology inference

Another way to build covers and use their nerves to exhibit the topological structure of data is to consider union of balls centered on the data points. In this section, we assume that $\mathbb{X}_n = \{x_0, \cdots, x_n\}$ is a subset of $\mathbb{R}^d$ sampled i.i.d. according to a probability measure μ with compact support $M \subset \mathbb{R}^d$. The general strategy to infer topological information about M from μ proceeds in two steps that are discussed in the following of this section:

1. $\mathbb{X}_n$ is covered by a union of balls of fixed radius centered on the x_i's. Under some regularity assumptions on M, one can relate the topology of this union of balls to the one of M;

2. From a practical and algorithmic perspective, topological features of M are inferred from the nerve of the union of balls, using the Nerve Theorem.